ANNUAL
REPORT

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Abstract

This report aims to better understand, through a detailed analysis, the activity of the company AEKI based on the exploitation of its sales data from 2014 to 2022. The objective is to understand its operating dynamics in order to propose a series of recommendations aimed at optimizing its performance.

Before performing any analysis, the data made available was cleaned and consolidated within a single Excel file before being integrated into Power BI. In particular, we corrected the erroneous values and replaced those that were missing. Once this step was completed, our analysis focused on six axes: temporal, geographical, customer, product, discount, logistics (delivery method). These dimensions were addressed through various visuals interacting with each other (linear graph, map, scatter plot, column bars charts). To these visuals are added four KPIs highlighting the following key elements: turnover, profit, profit margin, quantity sold.

The analysis carried out highlights two growth cycles (2014-2017 and 2020-2022), both separated by a slowdown phase in 2018-2019. Despite this observation, the profit margin remains stable, located around 12.5%, highlighting a solid economic model. Temporally, we note a marked seasonality. Indeed, sales are propelled at the end of the year (September, November, December) but fall sharply during the month of February. Geographically, the coastal areas, and more particularly California and New York, account for over a third of revenue, while other regions such as those in the Southeast show negative margins. On the product side, the *Canon imageCLASS 2200* establishes itself as a flagship product in terms of profit, while the *Cisco TelePresence System EX90* generates losses, particularly due to excessive discounts. The study of discounts shows that large reductions do not always imply significant sales, highlighting shortcomings in marketing. Finally, various logistical anomalies could be observed but depend strongly on the regions and shipping mode.

From these results, we can emphasize that the business model of AEKI remains solid despite some corrections to be made. Indeed, it is strongly recommended to review the promotion policy in order to maintain appropriate margins, adapt prices according to segments and regions, and provide marketing efforts during the months with high potential. In addition, better inventory management and optimization of logistics flows would strengthen the organization's results.

Introduction

This report presents a focused analysis of AEKI’s operational and financial performance over the period from 2014 to 2022, using a detailed look at the main sales and financial figures. Our main objective is twofold. First, we need to clearly map out how the company's business has generally progressed over these nine years. Second, the report aims to find the specific internal factors, like product sales or logistics, that have played a role in these results. Our goal is to give the organization practical recommendations based on the data.

To do this, the analysis is organized around key topics, including geographical differences, the impact of discount policies, product profitability and delivery efficiency. In this report, we describe the method we used for the data and provide an overview of the main financial figures. After that, the analysis moves on to examine sales trends and customer segments to find any regular patterns in buying behavior. The following sections look at regional performance, the promotion strategy, how much profit each product brings in, and the performance of delivery methods.

Each part of this report includes specific recommendations to improve the company’s performance and ensure its long-term profitability. The aim of this report is therefore not only to describe past developments, but also to provide useful elements for decision-making and the definition of future strategies.

Global analysis of KPIs

— TREND ANALYSIS —

Year	Total Sales (k$)	Sales variation	Total Profit (k$)	Profit variation	Profit Margin (%)	Total Quantity (units)	Quantity variation
2014	483,72		49,45		10,22	7572	
2015	470,53	-2,73%	61,62	24,61%	13,10	7979	5,38%
2016	608,5	29,32%	81,66	32,52%	13,42	9825	23,14%
2017	732,51	20,38%	93,26	14,21%	12,73	12000	22,14%
2019	483,72	-33,96%	49,45	-46,98%	10,22	7572	-36,90%
2020	470,53	-2,73%	61,62	24,61%	13,10	7979	5,38%
2021	608,5	29,32%	81,66	32,52%	13,42	9825	23,14%
2022	732,51	20,38%	93,26	14,21%	12,73	12000	22,14%

Between 2014 and 2022, we can observe two cycles with upward trends, separated by a period of slowdown:

▪ **Continuous growth (2014-2017):** During this period, the majority of our indicators are experiencing sustained growth. The year 2016 constitutes a record in the evolution of activities with an increase of **30%** in sales, **32%** in profit and **23%** in units sold.

▪ **Interruption and slowdown (2018-2019):** A break appears in 2018, the year for which we do not have data, interrupting the continuity of the time series. Having no values for our KPIs during this year, the changes in our indicators were measured between 2017 and 2019. From 2019, a notable decline in all KPIs is observed, suggesting a slowdown in activity. The values recorded for this period are similar to those of 2014.

▪ **Recovery and stabilization:** Between 2020 and 2022, a pattern similar to that observed during our first cycle of growth is emerging. In 2022, we are seeing levels comparable to those of 2017, reflecting a gradual recovery in activities and performance.

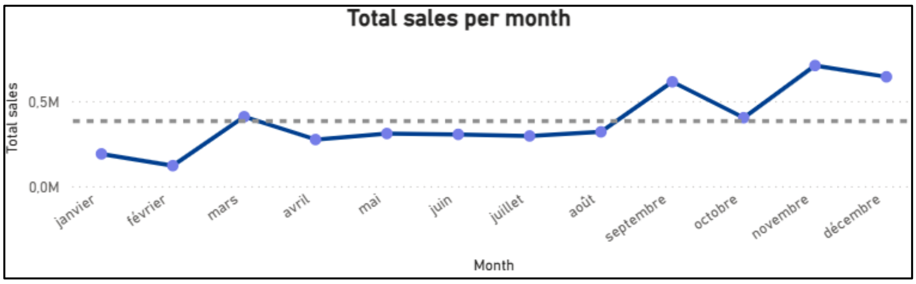
In the continuation of this report, these trends will be deepened by increasing the level of granularity of our analysis. Indeed, the following axes will be considered: temporal, client, product, geographical, logistics. This will allow us to identify the factors underlying performance variations.

— CONCLUSION —

The observed figures testify to a structural consistency of the company’s business model. Between 2014 and 2017, then between 2019 and 2022, the simultaneous increase in sales and profit indicators indicates that revenue growth was accompanied by controlled profitability. Throughout the period under review, the profit margin remained relatively stable, oscillating between 10% and 13%, including during 2019 constituting a phase of slowdown. Finally, the quantities sold follow a similar dynamic to that of turnover, highlighting that growth is essentially based on an increase in the volume sold.

Time-based sales analysis

— GLOBAL VIEW —



**Graph of the evolution of sales over all years*

Looking at the monthly turnover, we observe the following trends: February is generally the month with the lowest turnover followed by an increase in sales between February and March. A slight decrease is then observed between March and April, then sales stagnate from April to August. From August to September, sales are rising again but tend to decrease again in October. Finally, September, November and December are the months during which our sales are the highest.

— ANALYSIS BY TYPE OF CLIENT —

▪ **Individual customers** are responsible for **50% of sales** and therefore have a greater impact on overall variations. They tend to buy more in September, November, December but less in February.

▪ **Corporate clients** account for **31% of global turnover**. Their temporal consumption trends are the same as those of individual customers.

▪ **Home office clients** generate **19% of sales**. For 2014 and 2019, the turnover in the month of March was the highest. This trend is completely different for the other years where the best months are those of September, October, November and December.

— RECOMMENDATIONS —

▪ **Focus marketing efforts and promotional campaigns** before the end of the year, that is to say closer to the key months (September, November and December). The objective being to maximize performance during this period.

▪ **Stimulate demand** at the beginning of the year through stimulus campaigns. The objective being to smooth the seasonal effect and therefore better distribute the turnover over the entire year. It is very likely that the post-holiday season saturation affects consumers' budgets, and more specifically those of individual customers. Stimulating these buyers through promotional offers could therefore be a way to improve.

▪ **Adapt stock management** to seasonal trends. The objective is to anticipate sales peaks in order to avoid any breakdowns and additional operational costs.

Geographical analysis

— GLOBAL VIEW —

The map reveals a strong concentration of unit volume along the East and West Coasts. California (15K units; 13.2% margin) and New York (8,352 units; 20.8% margin) stand out as the company's commercial backbone, jointly representing more than one-third of national sales. Other major volume drivers are concentrated in the South and Midwest (e.g., Texas, Pennsylvania, Illinois).

In contrast, markets across the Central Plains (e.g., South Dakota, Nebraska, Oklahoma) record very low unit volumes but maintain excellent profitability per unit (margins often exceeding 16%), showing that margin efficiency can offset smaller market sizes. Wisconsin (20.9% margin) further illustrates this perfect balance - moderate volume, strong margin.

However, operational inefficiencies severely affect a subset of states. The South Atlantic and Mountain West regions, including Florida, Colorado, and North Carolina, show critically low or negative margins, highlighting that strong sales volumes often do not translate into profitability. The two loss-making states (Arkansas, North Carolina) represent the most urgent failures of unit cost management. Meanwhile, the Gray Zones (North Dakota, Montana, Wyoming, and Idaho) register no activity and represent untapped potential.

— RECOMMENDATIONS —

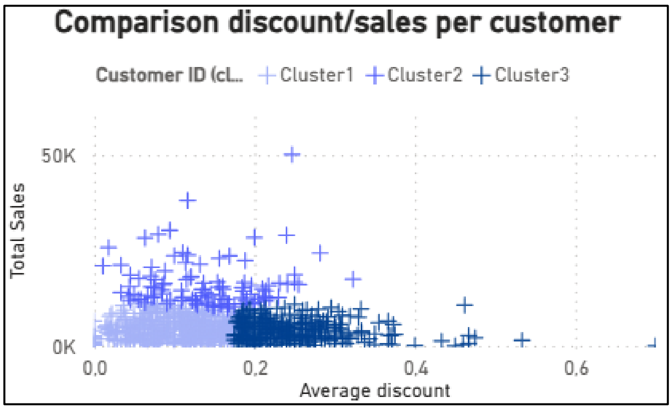
▪ **Profitability Restoration**: An immediate, comprehensive audit is paramount for the two loss-making states (Arkansas, North Carolina) and all low-margin, high-quantity markets (Florida, Colorado, etc.). The goal is to correct unit economics - pricing, costs, and discount policies - to ensure all units sold contribute positively.

▪ **Strategic Volume Growth**: Core Coastal Hubs (CA, NY) and major Southern/Midwestern markets (TX, PA, IL) must be consolidated. Concurrently, states across the Central Plains should be actively developed to increase profitable unit volume while strictly maintaining their existing high margins.

▪ **Zero-Volume Market Activation**: Gray Zone territories (ND, MT, WY, ID) must be explored via low-cost digital advertising and prospecting initiatives. This strategy aims to test customer acquisition potential and stimulate initial demand before committing to heavy structural or logistics investments.

Discount Analysis–sales

— GLOBAL VIEW —



This visual highlights three main clusters:

▪ **Cluster 1 (low discount, low/medium revenue):** Majority of customers with discounts below 20% and moderate purchase amounts.

▪ **Cluster 2 (low/medium discount, medium/high revenue):** Fewer customers but more profitable due to the application of moderate discounts on medium to high sales amounts.

▪ **Cluster 3 (medium/high discount, low/medium revenue):** Customers who frequently experience negative profits due to higher discounts (sometimes up to 40% or more) often applied for moderate to low purchase amounts.

Higher sales are not necessarily associated with customers who received the highest discounts, suggesting that a high discount is not consistently correlated to higher revenue. In contrast, the majority of customers generating significant sales are concentrated around a moderate average reduction.

— RECOMMENDATIONS —

▪ **Optimization of reduction policies:** It seems necessary to review the policy for allocating reductions and more particularly with regard to significant reductions (> 20%). The latter are, in part, applied to clients generating little turnover, which does not allow acceptable margins to be maintained.

▪ **Purchase incentive:** Encouraging customers belonging to cluster 3 to buy more would ensure suitable margins while ensuring sufficiently attractive discounts (customers with high price sensitivity). It is necessary to find a balance between customer acquisition and margin preservation.

Product Analysis

— GLOBAL VIEW —

The visual analysis of total sales vs. total costs (sales – profit) reveals the profit health of the top selling products. The “***Canon imageCLASS 2200 Advanced Copier***” stands out as the primary “cash cow”, with 40 units sold, generating over $123k in revenue and maintaining a solid 40% profit margin. This product needs a sustained stock and high visibility.

— ADDITIONAL ANALYSIS —

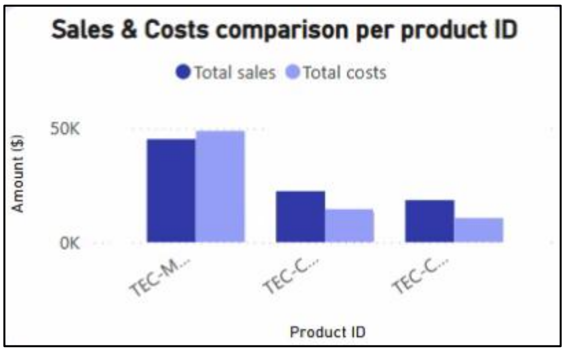
Temporal Analysis

As we go through top-selling products over time, an anomaly is exposed. During March 2014 and March 2019, sales volume went up due to the “***Cisco TelePresence System EX90 Videoconferencing Unit***” model (12 units in total). However, the associated costs were higher than the revenue generated for this specific product. This loss is likely driven by a 50% discount applied to the product. These sales effectively ran at an 8% negative margin, resulting in a net loss despite high sales amount.

Geographical Analysis

Analysis filtered by geography shows a significant difference in profitability. In the Central and Southern regions of the country, several top-selling products (including chairs, tables and video conferencing equipment) are generating costs that exceed sales amount.

Customer-based Products Analysis



When we analyzed the top-selling products, by Customer Segment, the “Home Office” category presents a risk. Unlike the other segments, the segment’s top-selling product generates a loss for the company.

The margin on the most popular Home Office product (in sales amount) is a -8% margin, because of the “***Cisco TelePresence System EX90 Videoconferencing Unit***” model.

— RECOMMENDATIONS —

▪ **Implement a hard floor on discounts for high-cost items** to prevent any sales from occurring below the cost of goods sold (COGS).

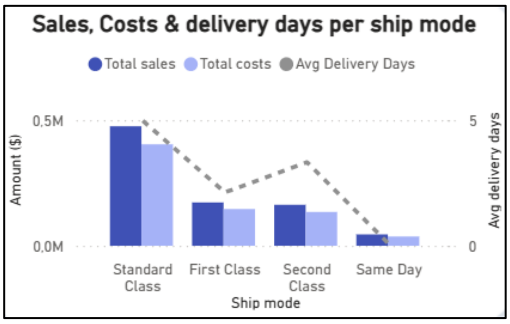
▪ Do an immediate **investigation into the logistics and costs** specific to the Central and Southern regions. If the costs are fixed, maybe implementing a regional price increase would be a solution to cover the expenses and generate a positive profit margin.

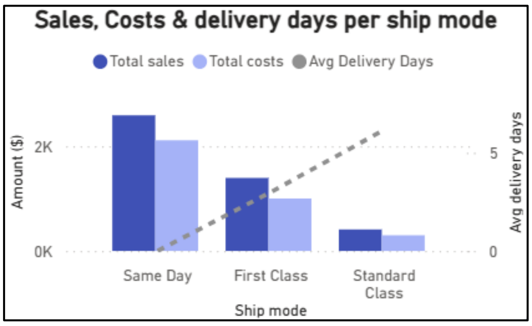
▪ **A re-evaluation of the pricing and discounts policy** for the Home Office segment. Consider separating the segment's pricing from the others to ensure a positive contribution to the overall profitability.

Ship mode analysis

— GLOBAL VIEW —

Cross-regional comparisons reveal consistent patterns: Standard Class is always the slowest, followed by Second, First, and Same Day - the quickest. The same order applies to sales volume, with Standard Class leading and Same Day marginal. However, certain client segments and regions deviate. For Home Office customers, First Class usually generates higher profits, though Second Class occasionally overtakes it (e.g., 2014, 2015, 2019, 2020). These clients generally value speed over cost.

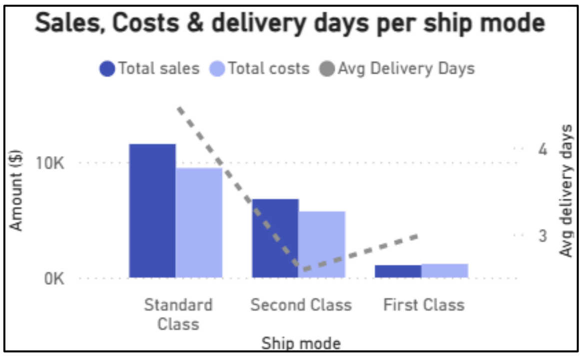




In the East, First Class is also favored, but some states differ: in Washington D.C., Same Day dominates, while in Rhode Island and Kansas, First Class leads clearly.

— DELIVERY ANOMALIES —

Several inconsistencies emerge in delivery times. In Maryland, First and Second Class durations are nearly identical despite price gaps. In New Hampshire, Second Class is faster on average than First. Elsewhere, Oregon shows longer Second than Standard Class times, and in Utah, First Class is slower than Second. These anomalies suggest logistical inefficiencies or regional inconsistencies.



Discrepancies likely stem from uneven logistics infrastructure (distance from distribution centers), carrier network limits (lack of dedicated First or Same Day capacity), and stock allocation issues (shipments from distant warehouses when local stock is low).

— RECOMMENDATIONS —

▪ Adopt region-targeted shipping: adjust default modes to local preferences and performance (e.g., prioritize First Class in Rhode Island and Kansas).

▪ Align pricing with real delivery efficiency.

▪ Audit logistics partners to detect bottlenecks and inconsistencies.

▪ Improve transparency by showing realistic regional delivery estimates to enhance satisfaction and conversion.

▪ Finally, use predictive models to recommend optimal shipping modes based on location, similar customer preferences, product type, and urgency.

Conclusion

This analysis confirms AEKI’s strong business model, supported by two growth cycles and a stable average profit margin of approximately 12.5%. However, our analysis showed some inefficiencies that, if left unaddressed, will impact the long-term profitability.

We have identified two critical areas driving these inefficiencies. First, high-cost products, like the Cisco TelePresence Unit, are running at a negative 8% profit margin, due to an aggressive discount that pushes sales below the costs of goods sold (COGS). Furthermore, the study of discounts shows that they are often misallocated to low-revenue customers.

Additionally, the geographical analysis highlights low or negative margins in the Central and South regions while the logistics analysis exposes anomalies where delivery costs and service levels are inconsistent across states.

To capitalize on AEKI’s strengths, we recommend acting across three pillars:

- Implement strict pricing governance (hard floor for discounts)
- Conduct a regional profitability audit (addressing costs in loss-making states)
- Optimize operational flow (resolving logistical inconsistencies and leveraging peak seasonality)

Addressing these critical leaks is important to secure AEKI’s long-term profitability.

Annexes

— DATA CLEANING PROCESS —

The initial dataset required several cleaning and transformation to ensure the data quality.

First, we created one Excel file from the csv and text files we had at the beginning. An incoherence was detected in the date format (pre and post 2018), and we corrected it with the “Text To Columns” tool in Excel to standardize the date format.

Secondly, two significant outliers were corrected based on the available data: the value 3311000 was adjusted to 33.11, and 14796 was corrected to 1.4796. We also identified empty cells (profit, sales, and discount) and replaced them with 0 to ensure accurate numerical aggregations.

Finally, following the confirmation of the data manager, we deleted the redundant elements in the IDs of products and customers. The clean data was then imported into Power BI, and a dedicated dim_date table was created to make the navigation through the dates easier.

— DESCRIPTION AND JUSTIFICATION OF THE VISUALS —

Key Performance Indicators (KPIs)

Key performance indicators (KPIs) allow to evaluate AEKI’s overall business performance. We have chosen four indicators: total sales and the total quantity sold, allowing to evaluate the level of activity, as well as the total profit and the profit margin, which measure the profitability of the activity. By choosing to show these indicators, the manager has a complete view of the company’s financial and commercial performance (notably by interacting with filters and other visuals).

Regarding the construction of these KPIs, it is relatively simple. Each of them is based on the use of the visual "card", in which we have dragged and dropped the four measurements from our fact_sales. The only particularity is the KPI of the profit margin, calculated by a measure dividing the total profit by the total sales, then multiplied by 100 to have a value expressed as a percentage.

```
Profit Margin (%) = DIVIDE(SUM(fact_sales[Profit]), SUM(fact_sales[Sales]))*100
```

Time-based sales analysis

The monthly granularity time analysis ensures the study of sales dynamics during the year. The interest of this type of analysis is to identify potential seasonal trends and periods of low or high activity. Unlike the global reading provided by the KPIs, this visual introduces a temporal dimension enriching our analysis.

Regarding the construction of this line graph, we used the visual 'line chart', with on the X-axis the months from the Month column of the dim_date table, and on the Y-axis the total sales from the sum of the sales from the fact_sales table. Markers (dots) have been added to make the visual clearer, as well as an average line to quickly visualize the months below or above the general trend.

Geographical analysis

The geographical visualization displays the distribution of total quantity sold across U.S. states. This map offers a clear, spatial overview of market activity and helps distinguish major hubs for unit volume from regions with lower transaction frequency.

Regarding the construction of this map, it was created using a Filled Map (Azure Map) visual, with the state field placed on the Location axis and the sum of quantity on the Color saturation axis. The color intensity directly reflects the volume of units shipped: the darker the blue shade, the higher the total quantity achieved in that region. A blue color gradient was applied based on quantity intervals to highlight sales differences between states.

Discount Analysis–sales

The discount-sales analysis highlights the relationship between the average level of discounts granted and the total turnover generated by each customer. The objective is to be able to determine the relevance of the discounts granted in relation to the sales made, making it possible to evaluate the effectiveness of our promotion policy. This visual draws our attention to a major strategic lever that is promotions and which greatly impacts the commercial dynamics of the organization.

Regarding the construction of this scatter plot, we simply used the average value of reductions (X axis) and the sales value (Y axis), both from the fact_sales table. The Customer ID serves as a reference value, completed by the Customer Name present in the info-bubbles section. Finally, an automatic clustering in three groups was generated by Power BI, allowing different customer profiles to be identified based on the level of discount granted and sales made.

Product analysis

Product analysis is essential for AEKI to compare sales and costs of top/bottom 'N' products. It allows to reveal the cash cows and profit leaks across various dimensions.

This section explains the construction of the dynamic visual, allowing the user to choose whether he wants to display the top or bottom N.

First, a base measure, "Total_sales", was created to sum the sales column. Then we created a "Sales_rank" measure, implemented using the RANKX function to assign a rank to each product based on "Total_sales".

```
Total_sales = SUM(fact_sales[Sales])
```

Secondly, a disconnected table was created to store the options "Top" and "Bottom" in a column named "Selection". This table's column was used to create a Slicer on the dashboard, allowing the user to choose the desired ranking direction. A new measure, Selected_value, was created using the SELECTEDVALUE function to capture whether the user selected "Top" or "Bottom" from the slicer.

```
Selected_value = SELECTEDVALUE('Table'[Selection])
```

The Sales_rank measure was then modified to dynamically adjust the ranking order based on the Selected_value ("Top" or "Bottom").

```
Sales_rank = IF([Selected_value] = "Top",RANKX(ALL(dim_product[Product ID]),
fact_sales[Total_sales],,DESC,Dense),RANKX(ALL(dim_product[Product
ID]),fact_sales[Total_sales],,ASC,Dense))
```

To allow the user to define how many items (e.g., Top 5, Bottom 10) they want to see, a Numeric Range Parameter was created (e.g., named NumericRange). This provides an input box for the user to enter the desired 'N' value. Next, the core filtering logic was implemented in the "Total sales" measure. This measure utilizes the Sales_rank and the value from the NumericRange parameter. It returns only the values of the Total_sales that falls within the Top/Bottom 'N'. Otherwise, it returns BLANK(). The exact same logic was used to create the measure with the associated costs.

```
Total sales =  VAR _selected =
SELECTEDVALUE(NumericRange[NumericRange]) RETURN
IF(fact_sales[Sales_rank]<= _selected, fact_sales[Total_sales],BLANK())
Total costs =  VAR Seuils_N =
SELECTEDVALUE('NumericRange'[NumericRange],3) VAR Product_rank =
[Sales_rank] RETURN IF(Product_rank<=Seuils_N, [Total_cost], BLANK())
```

Finally, the final visual was created. The category (e.g., Product_ID) was placed on the axis, and the “Total sales”(for Sales) and “Total costs” (for Costs) measures were added as values, resulting in two bars per category that dynamically filter based on the user's Top/Bottom and N value inputs.

Ship mode analysis

This visual examines average delivery duration (Order Date – Ship Date), total sales, and total costs for each shipping mode. That choice of visual was made because it lets us look at sales and costs by shipping mode to see if there’s a clear trend in profitability between them. It also helps us check the average delivery speed for each mode, to understand if the faster and more expensive ones are really more efficient in practice.

Regarding the construction of this visual, we use a combo chart combining clustered columns and a line chart. For the clustered columns part we placed on the X-axis the ship mode from the dim_ship table and on the Y-axis the sum of sales (dark blue) as well as the total cost (light blue), both coming from the fact_sales table. For the line chart we used a secondary Y-axis where we plotted the “average delivery days” (dotted grey line). We calculated the average delivery day with a measure that computes the average of the differences in day between the ship date (from the dim_ship table) and the order date (from the fact_sales table) for each order.

```
Avg Delivery Days = AVERAGEX(fact_sales,DATEDIFF(fact_sales[Order
Date],RELATED(dim_ship[Ship Date]),DAY))
```